

Yissum Research Intelligence Report — Week 31, 2026

5 high-potential paper(s) · Weekly · Week 31, 2026 · Generated July 27, 2026

This week's digest showcases cutting-edge advancements across diverse fields with significant commercial potential. Highlights include a novel metasurface platform enabling unprecedented thermal control for defense and electronics, and a breakthrough in AgriTech creating pest-resistant staple crops crucial for global food security. Further innovations feature smart gene therapies for neurological disorders, alongside new insights into epigenetic regulation poised to revolutionize diagnostics and drug discovery, including precision interventions for autism spectrum disorders. These developments represent robust opportunities for investors and industry partners seeking to engage with pioneering technologies.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Roy Maman** — Materials, Imaging
- **Osnat Malka** — AgriTech, Genomics, Synthetic Biology
- **Tawfeeq Shekh-Ahmad** — Drug Discovery, Neuroscience, Genomics, Clinical
- **Tommy Kaplan** — Genomics, Drug Discovery, Diagnostics, Software/AI
- **Haitham Amal** — Drug Discovery, Neuroscience, Proteomics, Genomics

ALL BRANCHES — RESEARCH HIGHLIGHTS

1. Metasurface platform for simultaneous and uncorrelated emissivity control over MWIR and LWIR spectral bands.

32/50

"Breakthrough metasurface independently controls thermal emissions, revolutionizing stealth, electronics, and advanced imaging applications."

Main researcher: Roy Maman · Roy.maman@mail.huji.ac.il.

Institute of Applied Physics, The Faculty of Science, The Center for Nanoscience and Nanotechnology, The Hebrew University of Jerusalem, Jerusalem, 91904, Israel.

Published in Light, science & applications · 2026 Jul 23

ABOUT THIS RESEARCH

Researchers have developed a metasurface capable of independently controlling how an object emits heat (thermal radiation) across two different infrared spectrums. This allows for the precise manipulation of thermal signatures and heat dissipation without one wavelength affecting the other.

COMMERCIAL ANGLE

High-value applications in stealth technology, advanced thermal management for electronics, and specialized non-invasive thermal imaging sensors.

COMMERCIALISATION METRICS

Novelty

7/10

The ability to achieve uncorrelated, simultaneous control over two distinct infrared bands (MWIR and LWIR) represents a significant functional leap over single-band or correlated metasurfaces, though it builds on existing plasmonic/dielectric resonator principles.

Commercial Potential

7/10

The ability to independently control MWIR and LWIR emissivity creates a high-value platform for thermal camouflage, energy-efficient coatings, and advanced sensing. While the technical application is clear, the score is tempered by the lack of provided data on scalability and manufacturing costs.

Market Size

7/10

The ability to independently control MWIR and LWIR emissivity is a platform-enabling capability with high value in defense (stealth/cloaking) and industrial thermal management, though it targets specialized high-end niches rather than a mass consumer market.

Tech Readiness

3/10

The paper describes the development of a metasurface platform for spectral control, which is characteristic of fundamental materials science and lab-stage proof-of-concept work. There is no evidence of integration into a commercial device, field testing, or scalable manufacturing processes.

IP Strength

8/10

The development of a platform for uncorrelated spectral control in two distinct bands represents a high-value 'platform-enabling' innovation with strong potential for broad patent claims across defense and industrial sensing.

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2. Creating resistance to the whitefly *Bemisia tabaci* in cassava through RNAi-mediated targeting of multiple insect metabolic processes. 31/50

"RNAi breakthrough creates whitefly-resistant cassava, offering a sustainable solution to protect vital global food crops."

Main researcher: Osnat Malka

Published · 2026-05

ABOUT THIS RESEARCH

Researchers developed a way to make cassava plants resistant to whiteflies, which are the primary carriers of devastating diseases in East Africa. They used RNA interference (RNAi) to target and disrupt essential metabolic processes within the insects.

COMMERCIAL ANGLE

Development of genetically engineered, pest-resistant crop seeds and RNAi-based biopesticides to reduce yield loss in global cassava production.

COMMERCIALISATION METRICS

Novelty

6/10

While targeting multiple metabolic pathways via RNAi is a robust strategy, host-induced gene silencing (HIGS) for pest control is a well-established technique; the novelty here lies primarily in the application to a specific crop-pest system rather than a fundamental scientific breakthrough.

Commercial Potential

7/10

The technology addresses a massive, billion-dollar agricultural pain point with a clear product path (transgenic resistant cultivars). However, the commercial potential is slightly tempered by the regulatory hurdles of GMOs in African markets and the necessity for multi-gene targeting to prevent insect resistance.

Market Size

8/10

The target market is vast and economically critical, with annual losses exceeding \$1.25 billion and direct impact on food security for millions in East Africa. The RNAi approach offers a scalable, platform-like solution to a 'super-abundant' pest affecting a primary global staple crop.

Tech Readiness

3/10

The research is in the early laboratory stage, focusing on the identification of candidate genes and the production of RNAi constructs, which precedes field testing and regulatory approval.

IP Strength

7/10

The use of multi-gene RNAi targeting specific metabolic pathways allows for strong composition-of-matter patents on the constructs and potential 'stacking' strategies to prevent resistance. However, defensibility is slightly tempered by the general nature of RNAi techniques and potential regulatory hurdles in agricultural biotech.

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3. Activity-dependent antioxidant gene therapy protects against epileptogenesis. 31/50

"Smart gene therapy prevents epilepsy post-injury by activating protective enzymes only when brain activity demands."

Main researcher: Tawfeeq Shekh-Ahmad · Tawfeeq.Shekh-Ahmad@mail.huji.ac.il.

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Published in Signal transduction and targeted therapy · 2026-06

ABOUT THIS RESEARCH

This research describes a gene therapy approach that uses natural brain activity to trigger the production of antioxidant enzymes. This mechanism aims to prevent the development of epilepsy following an initial brain injury.

COMMERCIAL ANGLE

Development of smart, responsive gene therapies for neurological disorders that activate only during specific physiological states to minimize side effects.

COMMERCIALISATION METRICS

Novelty

7/10

The work is highly innovative for shifting the therapeutic paradigm from reactive seizure control to proactive, activity-dependent gene modulation to prevent epileptogenesis. However, while the mechanism is sophisticated, antioxidant gene therapy is an established concept, making the novelty incremental rather than a complete disruption of biological principles.

Commercial Potential

7/10

The technology targets a high-unmet-need clinical indication (epileptogenesis) with a specific mechanism (antioxidant gene therapy), offering a clear path toward a specialized therapeutic product. However, the lack of abstract details regarding delivery vectors and scalability limits a higher score.

Market Size

7/10

The target market for epilepsy therapeutics is massive and high-value, representing a multi-billion dollar global neurology segment. However, the score is tempered by the specific niche of epileptogenesis prevention, which is a subset of the broader epilepsy market and faces complex clinical adoption hurdles.

Tech Readiness

3/10

The title suggests a fundamental mechanistic study using gene therapy in a disease model, which typically implies preclinical, lab-based proof-of-concept research rather than a developed therapeutic candidate.

IP Strength

7/10

The use of activity-dependent promoters for targeted gene therapy provides a strong basis for composition-of-matter and method-of-use patents, though defensibility depends on the novelty of the specific promoter/transgene combination.

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4. The genetic basis for DNA methylation variation across tissues and development.

30/50

"Comprehensive epigenetic map reveals genetic drivers of DNA methylation, unlocking new disease biomarkers and drug targets."

Main researcher: Tommy Kaplan · tommy@cs.huji.ac.il.

School of Computer Science and Engineering, The Hebrew University of Jerusalem, Jerusalem, Israel. tommy@cs.huji.ac.il; Dept. of Developmental Biology and Cancer Research, Institute for Medical Research Israel-Canada, Faculty of Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel. tommy@cs.huji.ac.il; Center for Computational Medicine, The Hebrew University of Jerusalem, Jerusalem, Israel. tommy@cs.huji.ac.il; Centre for Cancer Cell and Molecular Biology, Barts Cancer Institute, Queen Mary University of London, London, UK. tommy@cs.huji.ac.il; Centre for Epigenetics, Queen Mary University of London, London, UK. tommy@cs.huji.ac.il.

Published in Nature communications · 2026-04

ABOUT THIS RESEARCH

Researchers have mapped how specific genetic variations control DNA methylation patterns across different cell types and developmental stages in both mice and humans. This study identifies thousands of locations where DNA sequence changes drive epigenetic programming, linking genetics directly to gene regulation and disease risk.

COMMERCIAL ANGLE

This high-resolution atlas of sequence-dependent methylation can be leveraged to develop highly precise epigenetic biomarkers for early disease detection and to identify new regulatory targets for drug discovery.

COMMERCIALISATION METRICS

Novelty

9/10

The paper introduces a groundbreaking unifying framework that connects genetic sequence to epigenetic programming across species and developmental stages. It provides a massive, high-resolution atlas that transitions the field from descriptive epigenetics to a mechanistic understanding of how SNPs drive tissue-specific methylation.

Commercial Potential

6/10

While the atlas provides high-value foundational data for drug target discovery and disease mechanism modeling, it is a descriptive resource rather than a direct therapeutic or diagnostic product. Its commercial strength lies in its potential as a platform-enabling tool for biotech R&D rather than a standalone commercial offering.

Market Size

8/10

The atlas serves as a high-value, platform-enabling resource for drug discovery and functional genomics, potentially addressing the massive multi-billion dollar markets for precision medicine and target identification. By mapping the link between SNPs and methylation, it provides actionable data for interpreting disease-associated variants, a critical bottleneck in modern pharmaceutical R&D.

Tech Readiness

3/10

This is fundamental discovery science focused on mapping biological mechanisms and creating a descriptive atlas rather than developing a specific tool, diagnostic, or therapeutic intervention. While the data is high-value for downstream drug discovery, the research itself remains at the basic mechanistic understanding stage.

IP Strength

4/10

The paper is primarily fundamental discovery science that maps natural biological correlations rather than describing a proprietary tool, method, or molecule. While the resulting atlas is highly valuable for drug discovery, the underlying genetic and epigenetic associations themselves are natural phenomena and generally considered non-patentable.

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5. Phosphoproteomic profiling reveals reversal of dysregulated kinase signaling 28/50
by nitric oxide inhibition in the Shank3 mouse model of autism.

"Molecular targets identified for autism, offering precision pharmacological interventions for dysregulated brain signaling."

Main researcher: Haitham Amal · Haitham.amal@mail.huji.ac.il.

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Published in Scientific reports · 2026-06

ABOUT THIS RESEARCH

Researchers used phosphoproteomic profiling to map how nitric oxide inhibition restores normal signaling pathways in a mouse model of autism related to the SHANK3 gene. The study identifies specific protein phosphorylation changes that could explain how targeting nitric oxide synthase can reverse behavioral and synaptic symptoms.

COMMERCIAL ANGLE

The findings provide high-resolution molecular targets for the development of precision pharmacological interventions aimed at correcting dysregulated kinase signaling in autism spectrum disorders.

COMMERCIALISATION METRICS

Novelty 6/10

The study provides meaningful mechanistic insight by mapping the signaling pathways downstream of NO inhibition, but it follows a predictable biological logic rather than introducing a groundbreaking new paradigm or platform technology.

Commercial Potential 6/10

The study identifies a mechanistic target (nNOS inhibition) for a high-unmet-need indication, but the phosphoproteomic focus is primarily descriptive rather than proposing a novel, druggable modality or proprietary platform.

Market Size 9/10

Autism spectrum disorder represents a massive, global therapeutic market with significant unmet medical need and a high prevalence across diverse demographics. Targeting a fundamental neurobiological pathway like nNOS/phosphoproteomics offers potential for high-value, broad-spectrum clinical application.

Tech Readiness 3/10

The research is currently at the fundamental discovery stage, utilizing animal models (Shank3 mice) to identify mechanistic signaling pathways rather than testing a clinical candidate in humans. While it provides high-value biological validation, it lacks the preclinical development or pharmacological optimization required for higher TRL scores.

IP Strength 4/10

The paper describes fundamental mechanism-of-action (MoA) discovery rather than a novel composition of matter; while identifying signaling pathways is valuable for target validation, the use of nitric oxide inhibition is an established biological phenomenon that offers limited new patentable space.

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